ATTACHMENT A

PUBLIC INTEREST AND ENVIRONMENTAL STATEMENT

In Support of Petition for Environmental Review
Pursuant to 47 C.F.R. Section 1.1307(c)
ICFS File No. SAT-LOA-20260108-00016

This Attachment supplements the accompanying Petition for Environmental Review by addressing public-interest and environmental considerations that the present record requires the Commission to confront before grant. Factual assertions are drawn from the Application, other docket filings, published scientific literature, and publicly available government sources. Where this Attachment identifies disputed questions or inferences, it does so for record-preservation and agency-review purposes. To the extent this Attachment discusses scientific literature, regulatory implications, engineering feasibility, or policy consequences, it does so by citation, inference, and argument from the existing record, not as sworn expert testimony.

Record-document abbreviations used herein are as defined in the accompanying Petition for Environmental Review.

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I. THE DISPOSAL ARITHMETIC: WHAT THE RECORD DOES NOT ESTABLISH

The Application offers three disposal pathways: atmospheric reentry, Earth disposal orbits, and heliocentric disposal orbits. NAR at 5. The Technical Attachment states: "Disposal orbit altitudes have not been selected." TECH at A-12. The record does not bound propellant budgets for non-atmospheric disposal, does not state the expected disposal-method split, and does not demonstrate that either alternative pathway is operationally feasible at the described scale.

A five-year satellite lifecycle at one million satellites yields a reasonable illustrative steady-state replacement rate of approximately 200,000 satellites per year. This figure is offered because the Application provides no bounded lifecycle-throughput envelope. The Amazon LEO Petition to Deny independently derives this figure. Amazon LEO Petition at 10. At 200,000 reentries per year, the system would produce approximately 547 reentry events per day, each involving the complete ablation of a satellite composed of aerospace-grade materials and alloys the Application does not fully disclose.

The present record does not establish: the annualized replacement rate; the disposal-method split between atmospheric reentry and alternative pathways; the per-unit satellite mass and material profile for each hardware variant; the annual aggregate reentry mass; the material composition of reentering spacecraft; the cumulative atmospheric loading at the described cadence; or the launch cadence required for deployment and steady-state maintenance. The Commission cannot presently determine whether this lifecycle-throughput architecture constitutes a significant environmental effect.

II. LAUNCH CADENCE: WHAT THE APPLICANT'S OWN DESCRIPTION REQUIRES

The Applicant describes an illustrative scenario involving "launching 1 million tonnes per year of satellites." NAR at 4. The Application does not state which launch vehicles will service the constellation, what mix of vehicles will be used at each deployment phase, or the launch cadence required for initial deployment and steady-state maintenance. The following table is provided for illustrative purposes only, applying the Applicant's own publicly described tonnage to the payload capacities of its current and planned launch vehicles:

Payload per Launch	Launches per Year	Launches per Day
Starship (150 tonnes)	Approx. 6,667	Approx. 18
Starship (100 tonnes)	Approx. 10,000	Approx. 27
Falcon 9 reusable (15.6 tonnes)	Approx. 64,103	Approx. 176

These figures are derived from the Applicant's own description and are offered because the Application does not bound the launch cadence by vehicle type, deployment phase, or facility. The burden is on the Applicant to provide the Commission and the public with the vehicle mix, launch cadence, and facility allocation that will support the described system at each phase of deployment and at steady state. Without that disclosure, the Commission cannot evaluate whether any existing environmental review of the Applicant's launch facilities encompasses the cadence this system requires.

For context, 4,526 satellites were launched into orbit worldwide in 2025, a record year. Amazon LEO Petition at 10. The FAA's existing environmental reviews for SpaceX's Boca Chica launch site were prepared for launch frequencies substantially below these figures.

Furthermore, the Application's environmental certification on Form 312 states that the proposed system will have no significant environmental effect. The Application separately states that the system "will accept all harmful interference." The Commission should require the Applicant to clarify on the record whether its no-significant-impact certification encompasses the environmental effects of the launch cadence required to deploy and maintain the described system, or whether the certification addresses only on-orbit operations. For a novel system of this scale, the environmental effects of deployment and steady-state replacement are not collateral to the authorization. They are a structural consequence of it. The launch cadence is not incidental to the system's lifecycle; it is the system's lifecycle. If the Form 312 certification does not account for the cumulative environmental effects of the launches required to place and replace one million satellites on a five-year cycle, the certification is incomplete on its face. The burden is on the Applicant to provide the Commission with the data necessary to evaluate whether the environmental self-certification it executed covers the full scope of environmental consequences its own system description implies. For a system without precedent in Commission licensing history, that burden cannot be discharged by silence.

III. ARE NON-ATMOSPHERIC DISPOSAL PATHWAYS FEASIBLE AT SCALE?

The Commission should require the Applicant to answer the following question on the record before any grant:

Has the Commission evaluated whether heliocentric disposal is propellant-feasible at the described scale, or whether atmospheric burn-up is the only economically viable disposal pathway for this architecture?

Reaching heliocentric orbit from LEO altitudes of 500 to 2,000 km requires substantially greater delta-v than deorbiting to atmospheric reentry. If the propellant reserves required for heliocentric disposal are prohibitive at scale, then heliocentric disposal is not a real option. It is a narrative alternative that conceals the fact that atmospheric burn-up is the only operationally rational disposal pathway. The same analysis applies to Earth disposal orbits: if those orbits require propellant reserves that materially reduce the satellite's operational payload fraction, the economic incentive to choose atmospheric reentry is strong.

If the Applicant cannot demonstrate that non-atmospheric disposal pathways are operationally and economically feasible at scale, the Commission's environmental analysis must proceed on the assumption that atmospheric burn-up is the predominant disposal method.

IV. IRRETRIEVABILITY: FAST RADIO BURSTS AND NON-REPEATABLE SCIENCE

Fast Radio Bursts are transient, non-repeatable astrophysical events lasting milliseconds. An FRB obscured by a satellite's radio emission during that millisecond window is permanently lost. It cannot be re-observed. The event is over. The data is gone.

The AAS Petition demonstrates that aggregate emissions from one million satellites would exceed harmful interference thresholds in fourteen protected radio astronomy bands, including bands used for FRB detection. AAS Petition at 7, Table. Unintended electromagnetic radiation from satellite electronics is already detectable with current instruments. AAS Petition at 8 to 9. The proposed constellation would produce streak densities approximately 100 times higher than current satellite populations.

The same irretrievability applies to gravitational-wave electromagnetic counterparts. The 2017 GW170817 neutron-star merger demonstrated that rapid radio observation following a gravitational-wave trigger is essential for multi-messenger astronomy. Future merger events will occur during the operational lifetime of the proposed system.

The Commission's Table of Frequency Allocations provides that "all emissions are prohibited" in enumerated passive bands. 47 C.F.R. Section 2.106, footnote 5.340. These are non-discretionary protections. The present record does not demonstrate compliance at the proposed scale. The Commission cannot presently determine that non-discretionary passive-band protections will be maintained.

V. ATMOSPHERIC SCIENCE: THE EVIDENCE BEFORE THE COMMISSION

Study	Key Finding	Implication for This Application
Murphy et al., PNAS (2023)	Spacecraft-origin metals (Al, Li, Cu, Pb) detected in stratospheric aerosol above natural background. Approx. 10% of sulfuric acid particles >120 nm contain spacecraft-alloy metals.	Reentry contamination is already measurable at current constellation scales.
Wing et al., Comms. Earth & Env't (2026)	First time-resolved lidar measurement of lithium plume at 96 km altitude from	Individual reentry events produce traceable atmospheric signatures. At 200,000/year, this recurs 547 times daily.

	single Falcon 9 upper-stage reentry. Ablation begins at approx. 100 km.	
Maloney et al., JGR Atmos. (2025)	Radiative forcing from accumulated reentry alumina modeled at megaconstellation scales. Forcing becomes significant at projected cadences.	Cumulative atmospheric loading has climate-relevant consequences that scale with reentry throughput.
Maloney et al., JGR Atmos. (2022)	Black carbon from rocket launches has disproportionate radiative effect at stratospheric altitude.	6,667+ launches/year far exceeds any modeled scenario.
Wright, Boley & Byers, Space Policy (2026)	Existing megaconstellation plans yield 40% probability of ground casualties every five years.	One million satellites would elevate casualty risk to levels the Commission has previously found unacceptable.

The Commission holds no atmospheric jurisdiction. But its authorization of the launch and deployment cadence is the predicate federal act that enables the atmospheric loading. The Commission's public interest obligation requires it to consider whether authorizing this cadence would foreseeably contribute to environmental harms that the present record does not bound. To the extent the Commission concludes that atmospheric effects fall outside the scope of its NEPA obligations, it should identify on the record which federal agency holds jurisdiction over those effects and whether that agency has conducted environmental review at the scale described in the Application.

VI. NIGHT-SKY, RADIO-FREQUENCY, AND ORBITAL-ENVIRONMENT EFFECTS

A. Optical and Infrared

The AAS Petition presents simulations showing the proposed constellation would produce tens of thousands of naked-eye-visible satellites globally, with conditions where "there would be far more visible satellites than stars." AAS Petition at 3 to 5. Waste heat from computing nodes would raise the infrared background. AAS Petition at 6. The NSF-DOE Vera C. Rubin Observatory ($800M+ in federal investment) could become nearly inoperable for its designed mission. AAS Petition at 3 to 4. Artificial light at night could disrupt bird migration, pollination, and plant growth. AAS Petition at 9.

B. Radio Frequency

Aggregate emissions would exceed harmful interference thresholds in fourteen protected bands. Three interference pathways: Ka-band harmonics, aggregate spurious emissions, and unintended electromagnetic radiation from onboard electronics. UEMR from existing Starlink satellites is already observationally confirmed. AAS Petition at 6 to 9.

C. Orbital Environment

A 70-fold increase in active satellite population concentrated in narrow shells. Blue Origin raises concerns about capacity for additional entrants. Blue Origin Comments at 1 to 2. AAS cites modeling showing collision risks scaling nonlinearly with density, and ground-casualty probability already at 40% per five years for current megaconstellation plans. AAS Petition at 10 to 11.

VII. ALTERNATIVES TO ROUTINE ATMOSPHERIC DISPOSAL

The present application assumes, but does not justify, an architecture in which end-of-life spacecraft are routinely removed through atmospheric ablation. That design choice may reflect current industry practice, but industry practice is not the measure of NEPA adequacy, nor does it resolve whether reasonable alternatives or mitigation measures exist at the scale proposed here.

Petitioner does not assert that full satellite recovery or reuse is technically or economically feasible for this system on the current record. That is precisely the point. The record does not show whether the Applicant considered alternatives to routine design-for-demise disposal, including recovery, refurbishment, servicing, modular replacement, partial component recovery, longer-life architectures, or other approaches that could materially reduce reentry throughput and cumulative atmospheric loading. As developed in Section III, the record does not demonstrate that the non-atmospheric disposal pathways described in the Application are operationally or economically feasible at the described scale.

At a proposed scale of up to one million satellites, the question is no longer whether atmospheric burn-up is common. The question is whether the Commission may authorize a disposal model at the unprecedented scale described without requiring the Applicant to address whether less environmentally burdensome alternatives exist. That question belongs in any Environmental Assessment required in this proceeding.

VIII. WHAT THE COMMISSION CANNOT PRESENTLY DETERMINE: REQUIRED FINDINGS BEFORE GRANT

Before any grant, the Commission must make on-the-record findings addressing the following. The present record does not establish the inputs necessary for any of these determinations:

A. Passive-Band Compatibility

Whether the aggregate RF emissions of the proposed constellation are compatible with non-discretionary passive-band protections at fleet scale.

B. Atmospheric Loading

Whether the cumulative atmospheric loading from approximately 200,000 satellite reentries per year constitutes a significant environmental effect.

C. Disposal-Pathway Feasibility

Whether heliocentric and Earth disposal orbits are propellant-feasible and economically viable at scale, or whether atmospheric burn-up is the only viable disposal pathway.

D. Launch Cadence

Whether the launch cadence implied by the Application exceeds any existing FAA environmental baseline for the Applicant's launch facilities.

E. Scientific Infrastructure and Irretrievable Loss

Whether the degradation of federally funded astronomical infrastructure, including irreversible loss of non-repeatable transient observations, is consistent with the public interest.

F. Orbital Stewardship

Whether persistent occupation of the 500 to 2,000 km orbital shells by up to one million spacecraft belonging to a single operator is consistent with the Commission's stewardship obligations.

G. Unbounded Lifecycle Parameters

What the annualized steady-state replacement rate, disposal-method split, per-unit satellite mass and material profile for each hardware variant, aggregate reentry mass, and material composition are for the described system.

The Commission should state, in any order disposing of this Application, its findings on each of these dimensions and explain how it weighed irreversible resource loss against the benefits claimed by the Applicant. To the extent the Commission determines that any of these dimensions falls outside the scope of its authority or obligations, it should identify on the record which federal agency holds jurisdiction and whether that agency has conducted the relevant review at the scale described in the Application.

IX. PRESERVED SUPPLEMENTAL CONSIDERATIONS

The following considerations are preserved for the record and for any future administrative or judicial proceeding. They are not necessary to support the core relief requested in the Petition.

They are offered because the Commission's public interest analysis under 47 U.S.C. Section 309(a) extends beyond conventional spectrum management, and because the record should reflect the full scope of consequences this Application presents. To the extent any of the following considerations are deemed outside the scope of Section 1.1307(c), they are independently cognizable under the concurrent petition-to-deny classification requested in Section I.G of the Petition, and under the Commission's independent public-interest obligations before grant under 47 U.S.C. Section 309(a). The Commission may not grant an application without first determining that the public interest would be served, and that obligation is not confined to matters raised within the comment period.

A. The Spectrum as Statutory Trust

The Communications Act premises the Commission's authority on stewardship of a finite public resource. Section 301 states the purpose of the Act is "to maintain the control of the United States over all the channels of radio transmission." 47 U.S.C. Section 301. The Supreme Court has recognized that a licensee does not acquire a property right in spectrum and that regulatory control exists because private transmissions can preclude other uses. Red Lion Broadcasting Co. v. FCC, 395 U.S. 367 (1969).

The AAS Petition demonstrates that aggregate emissions from one million satellites would exceed harmful interference thresholds in fourteen protected radio astronomy bands. AAS Petition at 7, Table. The Commission's Table of Frequency Allocations provides that "all emissions are prohibited" in enumerated passive bands. 47 C.F.R. Section 2.106, footnote 5.340. These are non-discretionary protections. They exist because passive observation is destroyed at the moment of interference, not on average across a deployment period.

Where fleet-scale aggregate emissions foreseeably degrade the signal-to-noise ratio of protected passive bands, the Commission is not merely managing interference between competing users. It is adjudicating the permanent impairment of a resource it holds in trust. The stewardship obligation is sharpened when an applicant seeks novel waivers and a license at unprecedented scale while describing dynamic operational behavior that cannot be fully reduced to the Commission's required technical forms. This consideration is preserved.

B. Sky Access as Shared Commons

The observable sky is among the oldest and most universally shared public resources. Aurora borealis events, meteor showers, eclipses, and naked-eye astronomical phenomena constitute a cultural and educational heritage that predates recorded history. These observations are not static. Auroras in particular are transient, non-repeatable events whose documentation increasingly relies on high-sensitivity digital sensors. A satellite dimmed below naked-eye visibility remains a high-contrast streak that masks delicate transient luminosity in digital captures.

The AAS Petition presents simulations demonstrating that the proposed constellation would produce conditions where "there would be far more visible satellites than stars" for much of the globe, for much of the night, for much of the year. AAS Petition at 3 to 5. The Applicant's own Narrative describes sun-synchronous orbits ensuring sunlight exposure "up to more than 99% of the time." NAR at 3. This means the satellites would remain visible for nearly their entire orbital period. The AAS further documents that artificial light at night from this constellation could disrupt biological processes including bird migration, pollination, and plant growth. AAS Petition at 9.

The Commission's public interest standard encompasses the preservation of conditions under which the public experiences and transmits its cultural relationship to the sky. An aurora masked by satellite streaks in a digital capture is an event that cannot be replayed. Where licensing at this scale foreseeably impairs that relationship, the Commission should address the impairment on the record, not dismiss it as beyond regulatory cognizance. This consideration is preserved.

C. The Orbital Commons and the Public Trust Doctrine

The public trust doctrine holds that certain resources are inherent to the common welfare and that the sovereign's authority over them is fiduciary in character. The sovereign may grant private access but may not abdicate its supervisory interest or permit the resource to be substantially impaired without an articulated public benefit commensurate with the impairment. Illinois Central Railroad Co. v. Illinois, 146 U.S. 387, 453 (1892). The doctrine is not confined to navigable waters. It has been recognized in public lands held for conservation, Gould v. Greylock Reservation Commission, 350 Mass. 410 (1966); in ecological systems whose degradation threatens environmental values, National Audubon Society v. Superior Court, 33 Cal.3d 419, 434 (1983); and in resources whose common character the Supreme Court has acknowledged as a continuing matter of law, PPL Montana, LLC v. Montana, 565 U.S. 576, 603 (2012). The doctrine's Roman antecedent, res communes omnium, classified air, flowing water, the sea, and the seashore as resources common to all mankind and not subject to private appropriation. The orbital environment shares the structural attributes of a trust resource: it is finite, its carrying capacity is bounded by physics, it cannot be privately owned under the Outer Space Treaty, and its degradation is irreversible on human timescales.

The AAS Petition documents that the proposed constellation would increase the active satellite population by approximately 70-fold, concentrated in narrow orbital shells. AAS Petition at 10. Blue Origin notes that granting authority for a system of this scale raises questions about whether sufficient capacity would remain for additional entrants. Blue Origin Comments at 1 to 2. Collision risk increases nonlinearly with orbital density. AAS Petition at 10. Where a single private operator seeks authority to persistently occupy a shared physical domain to the potential exclusion or material impairment of other users, two questions are squarely presented: whether the authorizing agency holds a stewardship obligation over that domain, and if so, what standard of justification applies.

The Commission should state, in any order disposing of this Application, whether it considers the orbital shells between 500 and 2,000 km to be a shared resource subject to stewardship obligations, and if so, how persistent occupation by up to one million spacecraft belonging to a single operator is consistent with those obligations. See also Martin v. Waddell, 41 U.S. 367 (1842); Held v. State, No. CDV-2020-307, 2023 WL 5765550 (Mont. 1st Jud. Dist. Aug. 14, 2023); Joseph L. Sax, The Public Trust Doctrine in Natural Resource Law: Effective Judicial Intervention, 68 Mich. L. Rev. 471 (1970). This consideration is preserved.

At megaconstellation scale, the Commission should also consider whether future grant orders should incorporate a material-stewardship condition requiring that for every satellite committed to atmospheric ablation, the applicant demonstrate the removal or controlled recovery of an equivalent mass from the orbital environment. A "one up, one down" framework at scale would ensure that the net material burden on the atmosphere and the net object count in the orbital environment do not increase without bound over the operational life of the system. Such a framework would also create a regulatory basis for differentiating among applicants. An applicant that incorporates self-recovery, on-orbit refurbishment, modular component replacement, or extended-life architectures into a bounded lifecycle plan demonstrates a lower environmental cost per unit of authorized capacity than an applicant that describes only unbounded disposal pathways, however individually permissible those pathways may be under current rules. The Commission's public interest analysis under Section 309(a) is not limited to evaluating whether each discrete disposal event is individually compliant. It extends to whether the cumulative lifecycle architecture of the proposed system, taken as a whole, is consistent with responsible stewardship of the orbital and atmospheric environment at the scale authorized. Where two applicants seek comparable orbital resources, the Commission should state whether and how it weighs bounded, recovery-oriented lifecycle architectures against unbounded, ablation-dependent ones in its public interest determination. This consideration is preserved.

D. AI Governance and Classification

The Application describes infrastructure whose primary function, by the Applicant's own framing, is to provide "unprecedented computing capacity to power advanced artificial intelligence models and applications." NAR at 1. The technical architecture confirms this: optical inter-satellite links carry the primary workload; Ka-band radio is described as a "backup" for TT&C and critical mission phases. TECH at A-1. The system is primarily a compute network. The backup capability is communications.

The United States does not currently possess a comprehensive regulatory framework governing autonomous AI systems operating beyond territorial jurisdiction. If the system's AI models drift, produce emergent behaviors, or are repurposed in ways not contemplated at the time of authorization, the Commission's ability to exercise continuing supervision through RF-based licensing conditions alone is structurally limited. No existing agency is equipped to audit on-orbit inference logic, compel inspection of model behavior, or verify that the system operates within a defined scope.

The Commission should articulate, in any order disposing of this Application, the limiting principle that defines the boundary of Part 25 authorization for systems whose primary function is computation rather than communication. If no such limiting principle exists, the grant risks converting Part 25 into a general-purpose orbital permitting regime that the Communications Act was not designed to create. This consideration is preserved.

The federal government's evolving posture toward AI infrastructure is further illustrated by the Department of War's late February 2026 designation of a domestic AI company as a supply-chain risk to national security under 10 U.S.C. Section 3252, an action taken during the comment period for an application proposing to place unprecedented AI compute capacity in orbit. See Department of War, Statement on Supply Chain Risk Designation (Mar. 6, 2026); NPR, "Pentagon labels AI company Anthropic a supply chain risk 'effective immediately'" (Mar. 6, 2026). The Petitioner does not ask the Commission to adjudicate that separate federal action. The Petitioner observes that the Commission's public-interest analysis of an orbital AI infrastructure deployment should not proceed as though the broader federal government's assessment of AI infrastructure risks does not exist.

E. Security, Strategic Control, and Treaty Implications

A centralized, persistent orbital compute layer capable of supporting global data aggregation and AI inference is inherently dual-use and strategically significant. The Applicant describes a system marketed to "government users" and positioned as civilizational-scale infrastructure. NAR at 2 to 4. The public interest requires the Commission to consider whether a single private entity should hold operational control over such a system absent explicit congressional authorization, interagency coordination, and enforceable security conditions.

The February 2, 2026 merger between Space Exploration Technologies Corp. and xAI introduced new shareholders into the Applicant's corporate parent, including entities affiliated with foreign sovereign wealth funds. The Ownership Attachment filed with the Application predates this transaction and does not reflect the current shareholder base. The Commission's foreign-ownership review obligations under 47 U.S.C. Section 310(b) require an accurate ownership picture before grant. The nature of the proposed system -- persistent orbital compute and AI inference infrastructure marketed to government users and described as serving "billions of users globally" -- heightens the significance of the foreign-ownership and control inquiry. The Commission should require an updated ownership disclosure and complete its Section 310(b) analysis on the current ownership record before any grant.

The Fourth Amendment implications of persistent orbital architecture at this scale are activated by structure, not by intent. The Supreme Court has held that scale, persistence, and aggregation can transform data-collection infrastructure into a constitutionally governed domain regardless of whether each individual data point would be independently protected. Carpenter v. United States, 585 U.S. 296 (2018). A system described as providing "unprecedented computing capacity" for "billions of users globally" creates a persistent global data-aggregation and inference layer whose constitutional implications the Commission lacks delegated authority to adjudicate.

The Outer Space Treaty, Article VII, and the Liability Convention impose international liability on the United States as the launching state for damage caused by space objects. The aggregate contingent liability exposure from one million satellites with a five-year lifecycle has not been estimated in any document in the Application record. No interagency consultation with the Office of Management and Budget, the Congressional Budget Office, or the Treasury Department appears in the record. The Anti-Deficiency Act, 31 U.S.C. Section 1341, prohibits the United States from incurring obligations in advance of appropriations. The Petitioner does not assert that the Anti-Deficiency Act directly constrains the Commission's licensing authority. The Petitioner preserves the question of whether aggregate contingent liability exposure under the Outer Space Treaty, at the scale described in this Application, presents a fiscal dimension that should be addressed through interagency consultation before grant.

The Commission should identify, before any grant, the statutory basis for the resulting treasury exposure, or state that the question is beyond the scope of its determination. These considerations are preserved.

F. Instrumentation Replacement

The Commission's own spectrum policy has historically required users of shared bands to implement interference mitigation measures proportional to their impact. Environmental mitigation banking requires developers who destroy wetlands to create or restore equivalent habitat elsewhere. An instrumentation replacement obligation applies the same logic to the observational commons: if you degrade the layer, you replace the layer.

The scientific community currently relies on ground-based observatories, balloon-borne instruments, and a limited number of space-based platforms for radio astronomy, space-weather monitoring, and detection of transient celestial events such as Fast Radio Bursts. These facilities have received hundreds of millions of dollars in federal funding over decades. A million-satellite constellation operating in the 500 to 2,000 km altitude band would degrade the sensitivity and utility of these instruments through direct radio-frequency interference, increased sky brightness, and cumulative background noise. Certain celestial events, including Fast Radio Bursts and gravitational-wave electromagnetic counterparts, are non-repeatable. Interference with their detection constitutes irreparable harm to the scientific record. The loss is immediate and permanent.

The Commission should consider whether any grant order authorizing a system whose aggregate emissions foreseeably degrade the signal floor in protected passive bands, elevate sky brightness above natural baselines, or otherwise diminish the observational capacity of federally funded scientific infrastructure should include an instrumentation replacement condition requiring the applicant to deploy, at its own expense, scientific instrumentation sufficient to compensate for observational capacity lost to the authorized system's operation, with data access provided to federally funded research institutions at no cost. This obligation is not punitive. It is restorative. LEO-based scientific instruments deployed as part of the constellation would capture data with advantages ground-based facilities cannot match, including no atmospheric absorption, no terrestrial RFI, and continuous all-sky coverage. Such an obligation need not be solely a cost center for any affected applicant. Space-based observational data collected at LEO altitudes, calibrated and delivered at scale, represents a commercially valuable product. The Applicant could offer premium data services to research institutions, government agencies, and private observatories worldwide, creating an additional revenue stream that offsets any increase in per-station cost associated with hosting scientific instrumentation. The instrumentation replacement condition thus offers any applicant a path to compliance that is economically productive rather than purely burdensome, while ensuring that the observational commons degraded by the constellation's operation is restored rather than written off. This consideration is preserved.

* * *

All arguments and theories identified in this Attachment are preserved to the fullest extent available for any future administrative or judicial proceeding. No argument is waived by its summary treatment herein or by its non-development in any prior filing.

Respectfully submitted,

/s/ Nickolai G. Bakken

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